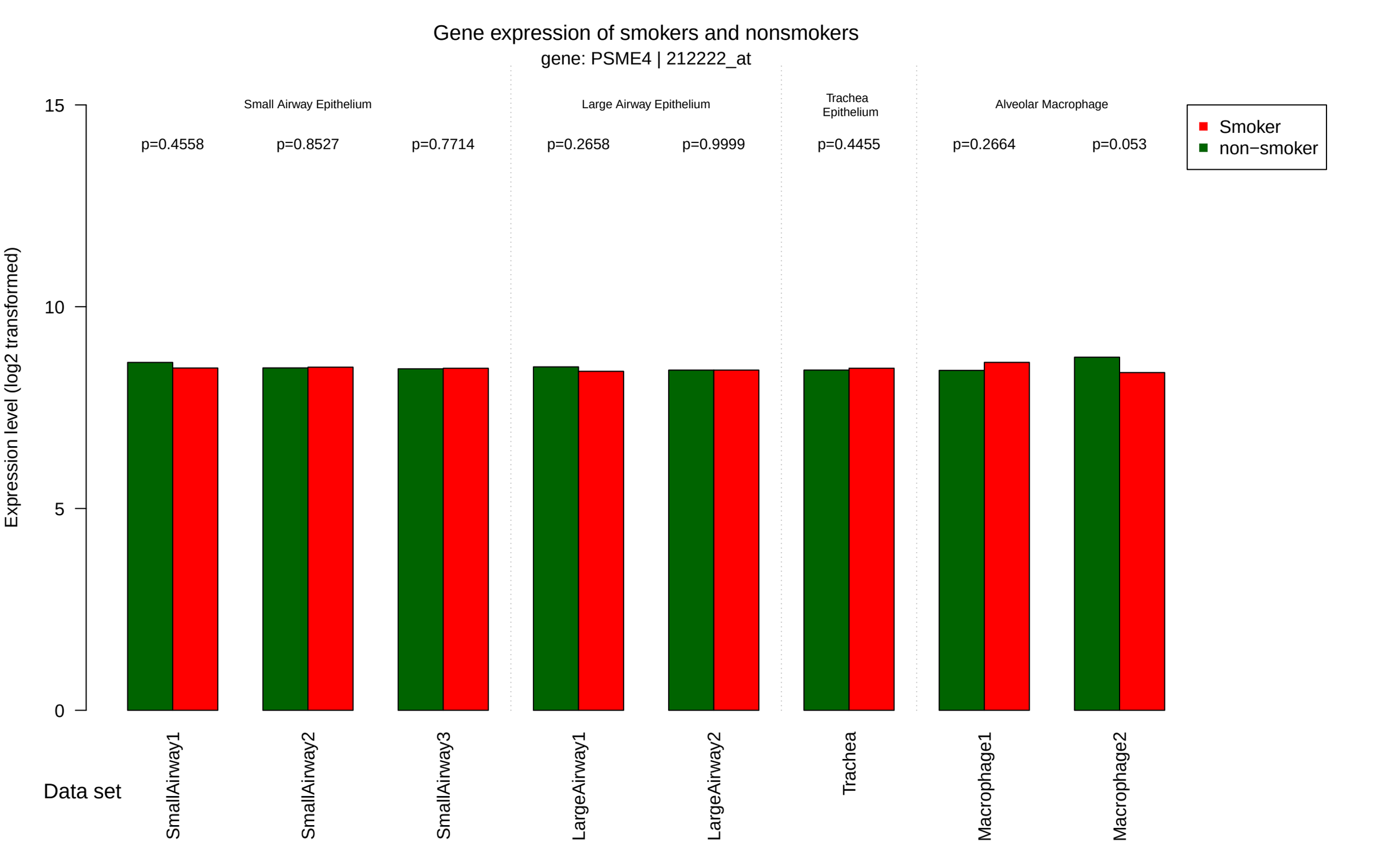
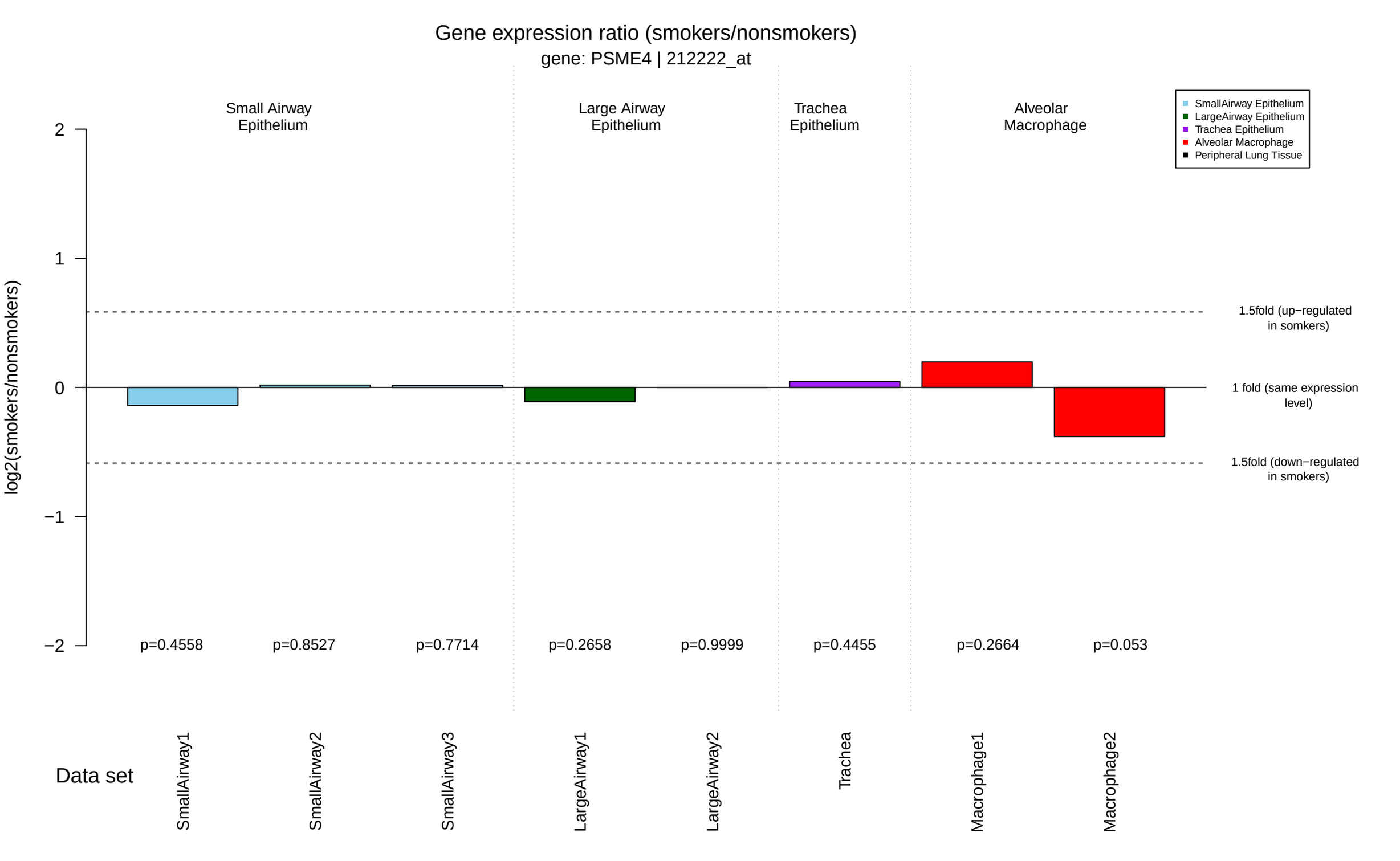
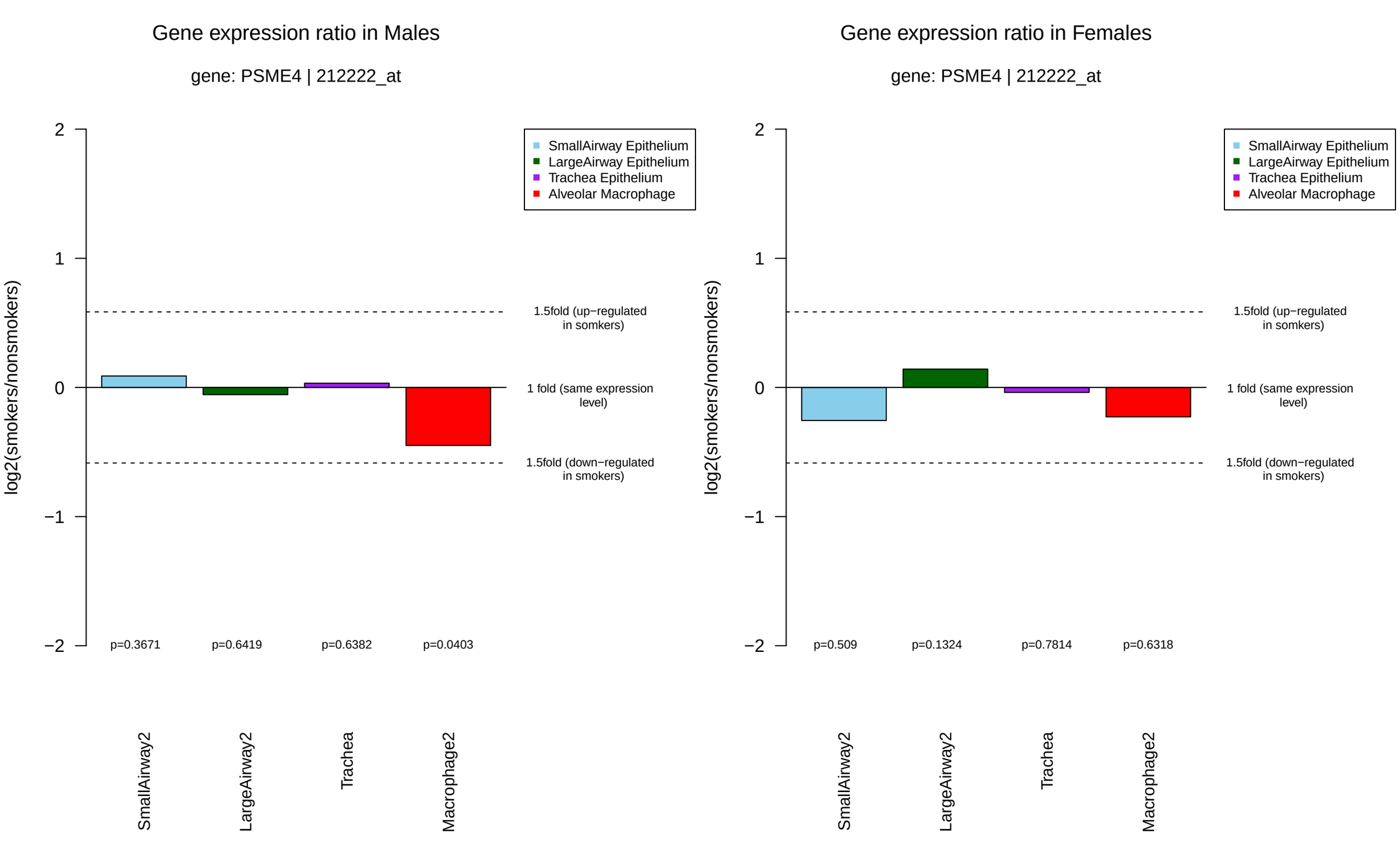
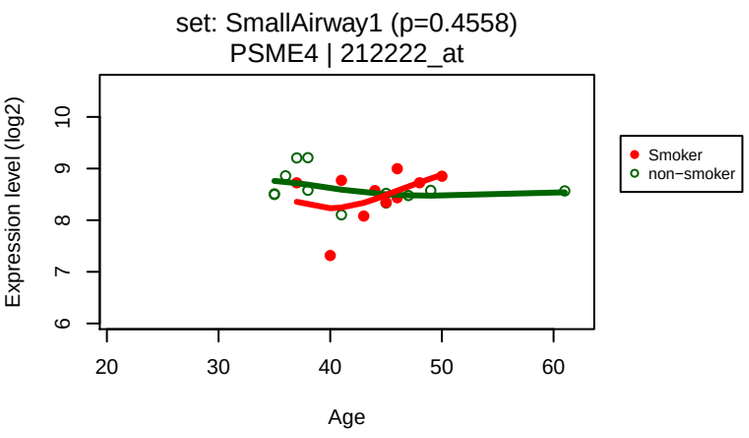
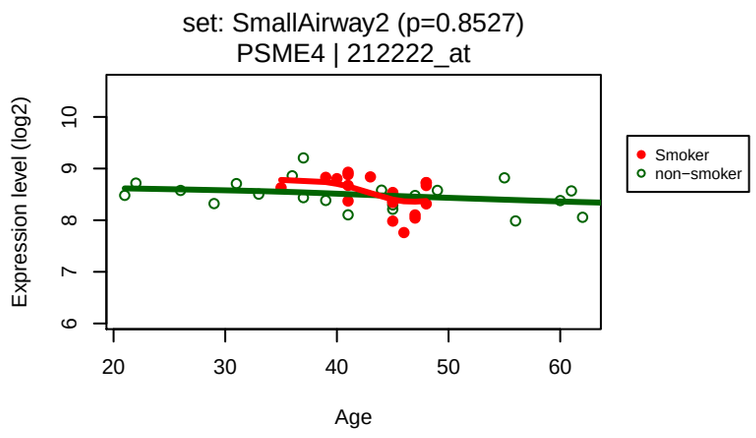
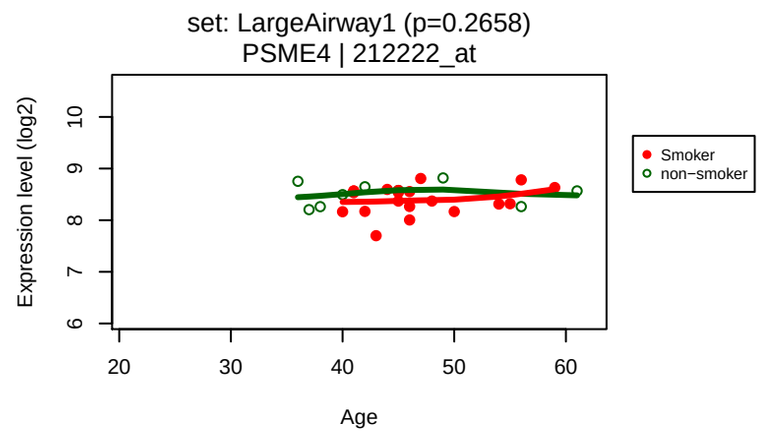

Gene expression of smokers and nonsmokers
gene: PSME4 | 212222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4558
p=0.8527
p=0.7714
p=0.2658
p=0.9999
p=0.4455
p=0.2664
p=0.053
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

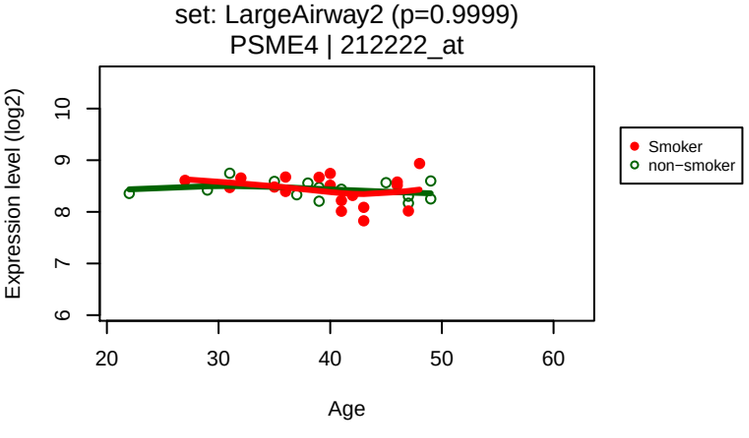

Gene expression ratio (smokers/nonsmokers)
gene: PSME4 | 212222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4558
p=0.8527
p=0.7714
p=0.2658
p=0.9999
p=0.4455
p=0.2664
p=0.053
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

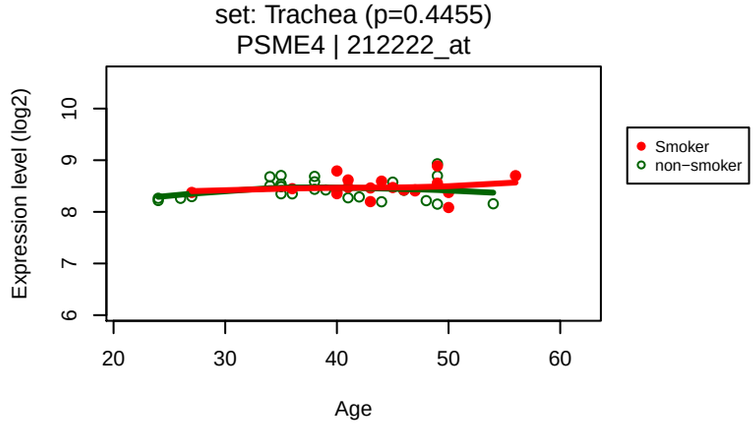

Gene expression ratio in Males
gene: PSME4 | 212222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3671
p=0.6419
p=0.6382
p=0.0403
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSME4 | 212222_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.509
p=0.1324
p=0.7814
p=0.6318
SmallAirway2
LargeAirway2
Trachea
Macrophage2

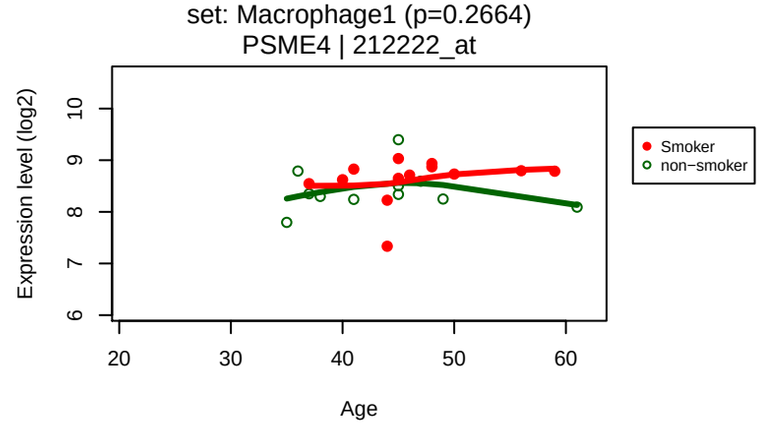

set: SmallAirway1 (p=0.4558)
PSME4 | 212222_at
Expression level (log2)
Age
Smoker
non−smoker

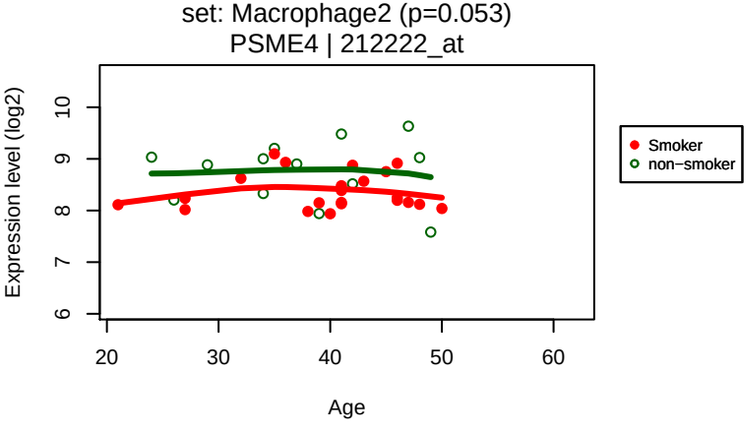

set: SmallAirway2 (p=0.8527)
PSME4 | 212222_at
Expression level (log2)
Age
Smoker
non−smoker

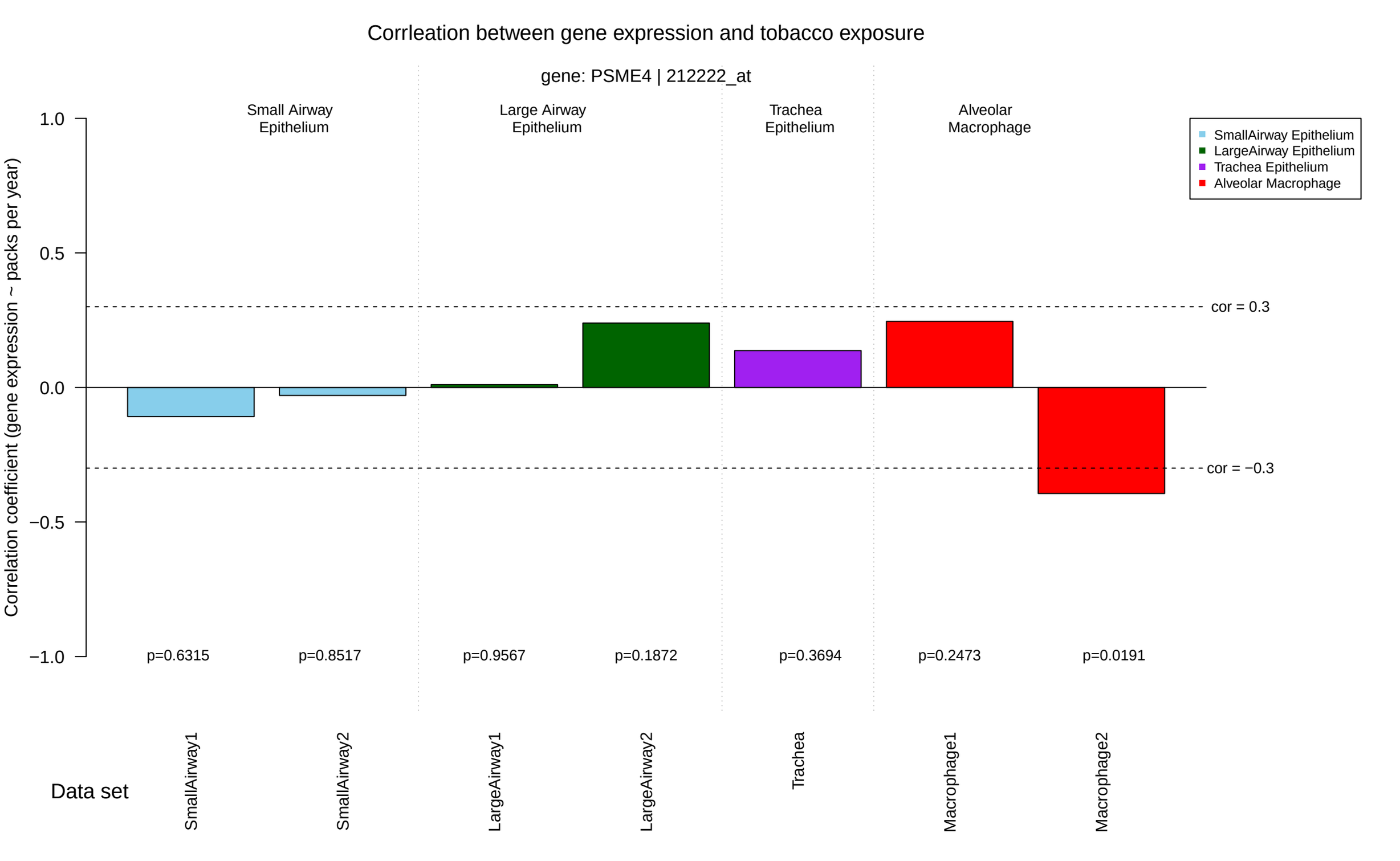

set: LargeAirway1 (p=0.2658)
PSME4 | 212222_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9999)
PSME4 | 212222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4455)
PSME4 | 212222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2664)
PSME4 | 212222_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.053)
PSME4 | 212222_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSME4 | 212222_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6315
p=0.8517
p=0.9567
p=0.1872
p=0.3694
p=0.2473
p=0.0191
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2